

Text-Only Large Language Models for Smart-Meter Energy-Theft Detection

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Abstract—Energy-theft detection on residential smart-meter data is usually treated as a supervised classification problem over engineered numeric features. This paper studies a different deployment question: whether hosted, general-purpose LLMs can consume the same meter evidence as text and produce both a theft label and a short explanation without task-specific training. We construct a text-only prompt that contains a clean reference week, a query day, temporal metadata, an attack catalogue, and a fixed output contract. We evaluate three hosted models—DeepSeek V4 Pro, Gemini Pro, and Gemini Flash—on 1,261 examples balanced across one original class and twelve attack classes. Gemini Pro obtains the strongest binary score, with balanced accuracy 0.976 and binary macro-F1 0.970. Gemini Flash is close on binary detection but weaker on attack-type identification, while DeepSeek V4 Pro trails both Gemini models on the multiclass task. The current evaluation also defines two explanation-scoring tracks, BERT-F1 and LLM-as-judge correctness, to be filled after scoring only the extracted `Attack_Behavior_Explanation` field. The results support a narrow claim: text-only LLMs are a plausible deployment baseline for smart-meter theft triage, but explanation quality still needs separate measurement.

Index Terms—energy theft detection, smart meters, non-technical losses, large language models, text-only prompting, explainable classification.

I. INTRODUCTION

Smart-meter deployments produce interval consumption records that can be used to detect Non-Technical Loss (NTL) from meter tampering, bypass, and false data injection. The common detection pipeline trains a classifier over numeric features and returns a scalar anomaly score. That design is efficient, but it gives limited support to an operator who must decide whether a flagged customer should be audited.

This paper focuses on a different contribution: deploying off-the-shelf LLMs as text-only classifiers for smart-meter theft detection. The model input is not a learned feature embedding. It is a structured prompt containing the reference-week readings, the query-day readings, summary statistics, temporal metadata, the attack definitions, and the required output format. The model is asked to decide whether the day is original or manipulated, identify the most likely attack type, and explain the manipulation behavior.

The deployment framing matters. We do not claim a new theft-detection architecture, nor do we train a domain-specific model. The question is whether existing hosted LLMs can be called as a zero-shot reasoning layer on the data already available to the utility. This is useful only if the models are scored like detectors, if the evaluation size is justified, and if explanations are treated as measurable outputs rather than assumed benefits.

We evaluate DeepSeek V4 Pro, Gemini Pro, and Gemini Flash on a 1,261-row text-only benchmark. Each class has 97 examples: one original class and twelve manipulated classes. Gemini Pro performs best in the current completed runs, reaching 0.976 balanced accuracy and 0.929 attack-type macro-F1. Gemini Flash remains strong on binary detection but drops to 0.763 attack-type macro-F1. DeepSeek V4 Pro reaches 0.892 balanced accuracy and 0.715 attack-type macro-F1.

The paper makes three contributions:

- 1) **Text-only LLM deployment protocol.** We define a prompt and parsing contract that uses only numeric smart-meter readings, metadata, and attack definitions.
- 2) **Model comparison on a fixed text benchmark.** We compare DeepSeek V4 Pro, Gemini Pro, and Gemini Flash on identical 1,261-row inputs using binary balanced accuracy and attack-type macro-F1.
- 3) **Explanation-evaluation hooks.** We separate label scoring from explanation scoring and reserve BERT-F1 and LLM-as-judge correctness for the extracted `Attack_Behavior_Explanation` field only.

II. RELATED WORK

Energy-theft detection has usually been studied with engineered features and supervised classifiers. CNN-based detectors reshape consumption history into structured daily arrays [1]; SVM ensembles and tree ensembles use hand-crafted statistics over load curves [2]; and deeper neural models extend the same pattern with learned sequence encoders [3]. These systems are optimized for label accuracy, not for producing explanations that an auditor can inspect.

Unsupervised and semi-supervised work treats theft as abnormal usage. Autoencoder and density-style detectors learn a normality model from clean consumption and rank later observations by reconstruction or local-density error [4], [5]. These methods reduce the need for attack labels, but they still return a score rather than a natural-language account of the suspected manipulation.

The LLM setting differs from both lines. The model receives the meter readings as text and emits a structured answer. This shifts the methodological question from feature learning to prompt design, output parsing, and explanation evaluation. The comparison in this paper therefore does not replace trained detectors; it establishes a deployable text-only baseline for models that can be called through hosted APIs.

III. PROPOSED APPROACH

Figure 1 summarises the pipeline. Each example consists of a clean seven-day reference week and one query day from the same consumer. The prompt includes the numeric 15-minute readings, per-day summary statistics, temporal fields, the attack catalogue, and the output contract. Ground-truth labels and filenames are not included.

The output contract has three fields. `Classification` is either `Original` or `Manipulated`. If manipulated, `Identified Attack` contains the most likely attack type, confidence, and reasoning. `Attack_Behavior_Explanation` describes how the consumption values are changed relative to the reference week, how magnitude and trend are affected, and which technical cause is plausible.

We treat the LLM as a deployable service rather than a trainable component. All calls are zero-shot. There are no few-shot examples, no fine-tuning, and no per-model prompt specialization. A parser converts the response into the binary label, attack label, and explanation field. Rows with parsing failure are excluded from label metrics unless a retry or offline parser update recovers the structured fields.

IV. EXPERIMENT SETUP AND EVALUATION METHODOLOGY

A. Data and Tasks

The evaluation set contains 1,261 examples. It is balanced at the class level: 97 original examples and 97 examples for each of twelve manipulated classes. The binary task maps original to the negative class and all attacks to the positive class. The multiclass task is attack-type identification over manipulated rows.

The attack catalogue covers full zeroing, uniform reduction, contiguous zero intervals, per-sample scaling, selected-period reduction, synthetic average-based values, cap-based manipulation, cut-off subtraction, flat average profiles, reversed usage profiles, lower-consuming-user substitution, and selective interval zeroing. The prompt includes the same definitions for all models.

B. Models

We report the three completed text-only runs:

- **DeepSeek V4 Pro**, using the high-reasoning setting and a truncation retry for length-limited rows.
- **Gemini Pro**, evaluated from the completed aggregate run in the 10 May result package.
- **Gemini Flash**, evaluated from the same 10 May result package.

Exploratory or partial runs are not included in the main tables. This keeps the reported comparison tied to complete, non-confounded result entries.

C. Research Questions

RQ1: How well do text-only LLMs detect manipulated consumption days in binary form?

RQ2: How well do the same models identify the attack type once manipulation is present?

RQ3: Is the 1,261-row evaluation large enough to make the binary comparison informative?

RQ4: How should explanation quality be scored without mixing the reasoning trace with the final explanation field?

D. Metrics

For binary detection we report true positives, false positives, false negatives, true negatives, precision, recall, specificity, balanced accuracy, and macro-F1. Balanced accuracy is the mean of attacked recall and original recall. It is the primary binary metric because the binary class distribution is intentionally imbalanced: 1,164 attacked rows and 97 original rows.

For multiclass attack identification we report attack-type macro-F1 over the manipulated classes. This is the metric available for all three completed runs. For DeepSeek V4 Pro we also compute the full thirteen-class macro-F1 internally, but the main comparison uses the attack-type macro-F1 to match the Gemini result files.

E. Sample-Size Check

Figure 2 gives a cumulative sample-size check. One evaluation sample group contains thirteen rows: one original row and one row for each manipulated class. We start with one group, compute balanced accuracy, add the next group, recompute balanced accuracy, and repeat until all 97 groups are included. This directly measures how the observed point estimate changes as examples are added. The DeepSeek curve is computed from row-level predictions; the current Gemini files contain aggregate counts only, so their completed scores are shown as final reference lines.

V. RESULTS AND DISCUSSION

A. Binary Detection

Table I reports binary detection. Gemini Pro is the strongest model on this task. It has only four false positives and seven false negatives, giving balanced accuracy 0.976 and macro-F1 0.970. Gemini Flash remains close on attacked recall but has more false positives on original rows, reducing balanced accuracy to 0.945. DeepSeek V4 Pro has the largest gap between attacked recall and original recall; it misses 35 attacks and incorrectly flags 18 original examples.

The binary result is strong but should not be overstated. The benchmark is class-balanced by construction at the attack-type level and has a much higher manipulated-row proportion than a deployed utility stream. In deployment, false positives would be more expensive because the operational theft base rate is lower. Specificity therefore matters: Gemini Pro is strongest on the original class, while DeepSeek V4 Pro is the weakest of the three on original recall.

Text-only LLM evaluation pipeline

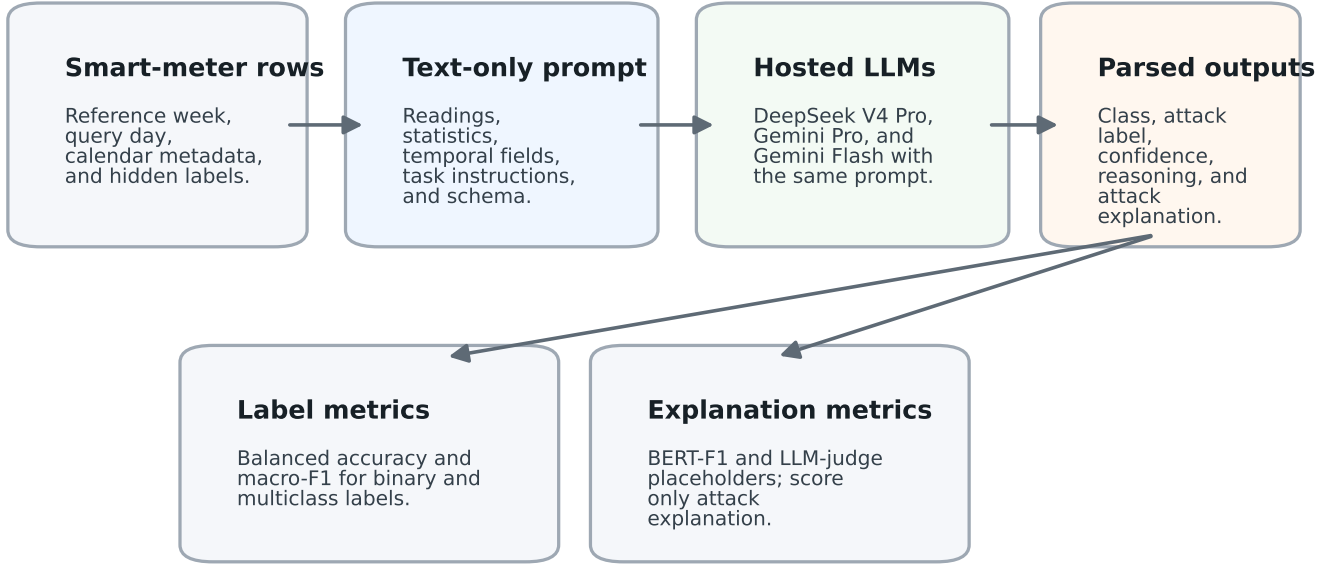


Fig. 1. Text-only LLM evaluation pipeline. Each call receives numeric smart-meter evidence and a fixed output contract. The parsed output is scored for labels first; explanation scoring is reserved for separate BERT-F1 and LLM-as-judge analyses.

TABLE I
BINARY DETECTION ON THE 1,261-ROW TEXT-ONLY EVALUATION SET. ORIGINAL IS THE NEGATIVE CLASS; ALL ATTACKS ARE THE POSITIVE CLASS.

Model	TP	FP	FN	TN	Bal. Acc.	Macro-F1	Prec.	Recall	Spec.
DeepSeek V4 Pro	1129	18	35	79	0.892	0.863	0.984	0.970	0.814
Gemini Pro	1157	4	7	93	0.976	0.970	0.997	0.994	0.959
Gemini Flash	1155	10	9	87	0.945	0.947	0.991	0.992	0.897

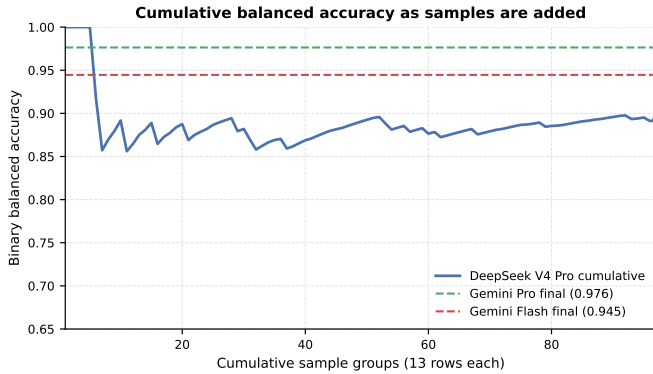


Fig. 2. Cumulative binary balanced accuracy as evaluation samples are added. Each sample group contributes thirteen rows. The DeepSeek curve is recomputed after each added group; Gemini Pro and Gemini Flash are shown as final aggregate reference scores because row-level Gemini outputs are not available in the current result package.

TABLE II
ATTACK-TYPE IDENTIFICATION OVER MANIPULATED CLASSES.

Model	Macro Prec.	Macro Recall	Macro-F1
DeepSeek V4 Pro	0.780	0.722	0.715
Gemini Pro	0.944	0.927	0.929
Gemini Flash	0.855	0.772	0.763

B. Attack-Type Identification

Table II and Figure 3 report attack-type macro metrics. Gemini Pro is again strongest, with attack-type macro-F1 0.929. Gemini Flash obtains 0.763 macro-F1, which is acceptable as a binary triage model but weaker for explaining the specific manipulation type. DeepSeek V4 Pro obtains 0.715 attack-type macro-F1 and 0.729 full thirteen-class accuracy in the row-level analysis.

The attack-type result is the more important stress test. A model can flag a day as abnormal while still confusing two low-magnitude manipulation mechanisms. The current results suggest that Gemini Pro is the only run that is consistently

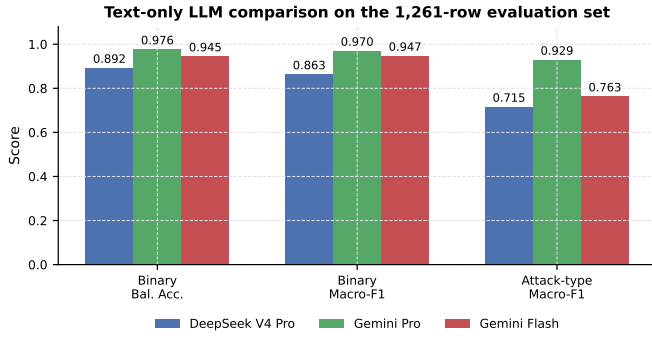


Fig. 3. Comparison of the three completed text-only LLM runs. Gemini Pro is best on binary balanced accuracy, binary macro-F1, and attack-type macro-F1.

useful for both detection and attack identification. Gemini Flash is plausible for low-cost binary screening, but its attack-type explanations require more checking. DeepSeek V4 Pro is weaker on the structured label task despite its reasoning mode.

C. False-Positive Case Study

The final paper should include one original example that all three models classify as manipulated, with a short summary of each model’s explanation. The current Gemini result package contains aggregate confusion counts but not row identifiers or raw explanations, so the shared false-positive row cannot be verified from the available files. Table III keeps the exact structure for this analysis and fills the DeepSeek row from an observed false positive.

This case study should be read as a qualitative diagnostic, not as an additional metric. Its purpose is to show whether the models make the same error for the same reason. If all three models flag the same original day because ordinary low-usage intervals resemble an on-off attack, then the failure is in the prompt’s distinction between rare normal behavior and manipulation. If the explanations differ, then the same label error may hide different model-specific reasoning paths.

D. Explanation Scoring Placeholders

The label metrics above do not measure explanation quality. We therefore reserve two explanation metrics for a later scoring pass:

Separating explanation scoring from label scoring is necessary because a correct attack label can have a weak explanation, and a plausible explanation can accompany the wrong label. The current manuscript therefore reports label results as completed and marks explanation correctness as pending.

E. Limitations

The evaluation has three main limitations. First, the attack catalogue is synthetic and derived from prior smart-meter theft studies; real theft may not match these transformations. Second, the benchmark is balanced by attack class and is not an operational base-rate simulation. Third, the Gemini files

available for this revision are aggregate metric files, so row-level qualitative analysis is incomplete until raw outputs are linked.

These limitations do not invalidate the comparison, but they narrow the claim. The present result is evidence that text-only LLMs can be deployed as a structured zero-shot baseline on numeric smart-meter prompts. It is not evidence that they are ready to replace trained detectors in a utility audit pipeline.

VI. CONCLUSION

This draft reframes the study around text-only LLM deployment for energy-theft detection. On the completed 1,261-row benchmark, Gemini Pro is the strongest model, with binary balanced accuracy 0.976 and attack-type macro-F1 0.929. Gemini Flash is close for binary triage but weaker for attack identification. DeepSeek V4 Pro is usable but trails both Gemini runs on the main metrics. The next required step is explanation evaluation: score only the extracted `Attack_Behavior_Explanation` field with BERT-F1 and an LLM-as-judge protocol, then add a verified common false-positive case once row-level Gemini outputs are available.

REFERENCES

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TABLE III
FALSE-POSITIVE REASONING CASE STUDY. THE SHARED ROW IS PENDING ROW-LEVEL GEMINI OUTPUTS; THE DEEPSEEK ROW SHOWS THE INTENDED LEVEL OF SUMMARY.

Model	Predicted label	Reasoning summary
DeepSeek V4 Pro	Attack 2	On an original row for CONS320, the model interpreted a contiguous block of low or zero readings as an on-off manipulation and treated the rest of the day as otherwise normal.
Gemini Pro	[PENDING]	Fill with the row-level explanation once the common false-positive row is aligned.
Gemini Flash	[PENDING]	Fill with the row-level explanation once the common false-positive row is aligned.

TABLE IV
PLANNED EXPLANATION-QUALITY METRICS. ONLY
ATTACK_BEHAVIOR_EXPLANATION IS SCORED.

Metric	Planned use
BERT-F1	Compare the extracted attack-behavior explanation with the catalogue description of the predicted attack. Do not score the full reasoning trace.
LLM-as-judge	Ask an independent judge model whether the explanation correctly describes the manipulation mechanism, magnitude/trend impact, and plausible technical cause.